DP: TASK-02
Feature Scaling and Comparative Analysis of Fitness Dataset using StandardScaler and Min-Max Scaler

1. Objective

The objective of this task is to prepare a fitness tracking dataset for machine learning by applying feature scaling techniques. The dataset contains numerical features collected from different devices such as steps, calories burned, and workout duration. Since these features are measured on different scales, feature scaling is necessary to improve machine learning model performance.

2. Introduction

Feature scaling helps normalize feature ranges, improve model accuracy, speed up convergence, and prevent larger-value features from dominating smaller ones. This report focuses on StandardScaler and MinMaxScaler.

3. Dataset Description

Dataset used: fitness_activity_100+_preprocess.csv. The dataset contains fitness activity records collected from users.

4. Tools and Libraries Used

Google Colab, Pandas, Scikit-learn, StandardScaler, and MinMaxScaler were used for preprocessing and analysis.

5. Importing Required Libraries

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, MinMaxScaler
```

6. Loading the Dataset

```
df = pd.read_csv('fitness_activity_100+_preprocess.csv')
df.head()
```

7. Dataset Information and Summary Statistics

```
print(df.shape)
print(df.info())
print(df.describe())
```

8. Selecting Numerical Columns

numerical_columns = df.select_dtypes(include=['int64', 'float64']).columns

9. Standard Scaling using StandardScaler

StandardScaler standardizes the data so that the mean becomes 0 and standard deviation becomes 1.

Formula:
$z = (x - \mu) / \sigma$

10. Code for Standard Scaling

```
standard_scaler = StandardScaler()
standard_scaled_data = standard_scaler.fit_transform(df[numerical_columns])
```

11. Min-Max Scaling using MinMaxScaler

MinMaxScaler transforms the data into a fixed range between 0 and 1.

Formula:
$x' = (x - xmin) / (xmax - xmin)$

12. Code for Min-Max Scaling

```
minmax_scaler = MinMaxScaler()
minmax_scaled_data = minmax_scaler.fit_transform(df[numerical_columns])
```

13. Comparison of StandardScaler and MinMaxScaler

StandardScaler produces data centered around zero with unit variance, while MinMaxScaler scales values between 0 and 1.

14. Applications of StandardScaler

Used in Logistic Regression, SVM, KNN, PCA, and Linear Regression.

15. Applications of MinMaxScaler

Used in Neural Networks, Deep Learning, and Image Processing.

16. Interpretation and Observations

StandardScaler produces negative values and standardizes distributions. MinMaxScaler preserves original distribution shape while normalizing values.

17. Conclusion

```
Both scaling techniques successfully transformed the dataset into machine-learning-friendly formats. The choice
```

18. References

Python Documentation
Scikit-learn Documentation
Pandas Documentation
Google Colab Documentation

Comparison Table

Feature	StandardScaler	MinMaxScaler
Scaling Method	Standardization	Normalization
Output Range	No fixed range	0 to 1
Mean	0	Not fixed
Standard Deviation	1	Not fixed
Negative Values	Possible	Not possible
Effect of Outliers	Less sensitive	More sensitive